

BEJAIA SOUMMAM ABANE RAMDANE, Algeria

WMO: 604020

Lat: 36.712N

Lon: 5.070E

Elev: 20

StdP: 14.69

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.6	42.4	33.9	28.6	50.4	35.9	30.9	49.2	23.7	57.9	22.0	56.4	8.9	240	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 15.1	(c) 91.0	(d) 73.7	(e) 87.8	(f) 75.4	(g) 85.9	(h) 75.5	(i) 79.6	(j) 85.7	(k) 78.3	(l) 84.5	(m) 77.2	(n) 83.5	(o) 9.2	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	142.8	84.0	76.5	138.3	83.3	75.2	132.1	82.2	43.0	85.8	41.7	84.8	40.5	83.6	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.6	(b) 16.8	(c) 14.7	(e) 35.9	(f) 102.2	(g) 3.7	(h) 5.0	(i) 33.3	(j) 105.8	(k) 31.1	(l) 108.8	(m) 29.0	(n) 111.6	(o) 26.3	(p) 115.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	53.8	53.8	57.1	60.5	65.6	72.0	77.3	78.8	74.5	69.6	61.4	56.0		
	DBStd	9.88	4.56	5.00	4.83	4.22	4.36	4.24	3.71	3.78	4.17	5.00	5.29	4.59		
	HDD50	41	15	17	3	0	0	0	0	0	0	0	1	5		
	HDD65	1527	348	315	250	147	41	1	0	0	0	11	132	282		
	CDD50	5552	132	123	223	315	482	660	845	892	736	609	343	192		
	CDD65	1566	1	2	5	12	58	212	380	427	286	155	24	4		
	CDH74	12156	3	13	39	58	211	1266	3455	4162	2063	777	99	10		
(13)	CDH80	3381	0	1	10	14	48	230	1002	1428	467	166	14	1		
(14)	Wind	WSAvg	6.9	7.9	7.5	7.0	6.6	6.1	6.2	6.3	6.2	6.2	6.8	8.3		
(15) Precipitation	PrecAvg	29.0	4.2	3.4	2.4	2.5	1.9	0.7	0.2	0.5	2.0	3.1	3.9	4.5		
	PrecMax	40.2	10.9	11.0	6.4	7.3	5.6	2.3	2.6	2.8	7.0	7.5	7.7	10.5		
	PrecMin	15.2	0.5	0.4	0.4	0.4	0.1	0.1	0.0	0.0	0.3	0.6	1.1	0.4		
	PrecStd	5.9	2.4	2.3	1.5	1.5	1.3	0.6	0.5	0.6	1.4	2.0	2.0	2.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.0	75.3	79.1	80.5	85.9	91.2	95.7	97.7	91.7	91.5	82.0	74.3		
		MCWB	58.0	59.3	61.3	62.1	69.3	72.9	73.2	73.7	72.9	69.1	63.1	59.6		
	2%	DB	67.7	68.5	72.2	74.0	78.5	85.0	89.2	90.7	86.3	83.9	76.8	69.8		
		MCWB	56.4	57.2	59.4	61.6	67.3	72.9	74.7	75.2	74.1	68.5	62.4	57.8		
	5%	DB	64.6	64.9	68.4	71.1	75.5	82.3	86.2	88.0	84.3	80.5	73.3	66.6		
		MCWB	55.0	55.9	59.1	61.7	66.9	72.9	76.1	76.8	74.1	69.9	62.2	56.8		
	10%	DB	62.2	62.3	66.0	68.5	73.5	80.1	84.5	86.1	82.5	78.1	70.1	64.2		
		MCWB	54.2	55.1	58.4	61.1	66.2	72.1	75.6	76.7	73.5	69.1	61.0	55.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.8	61.1	63.6	66.5	72.0	77.4	80.8	81.3	79.3	75.3	68.7	62.5		
		MCDB	68.2	69.3	73.0	73.1	79.5	83.8	86.3	87.1	84.7	80.9	75.0	69.2		
	2%	WB	58.1	58.9	61.7	64.5	69.7	75.7	79.2	80.0	77.2	73.2	65.6	59.8		
		MCDB	65.0	66.1	68.6	70.4	75.7	81.9	85.0	86.3	83.1	79.1	72.1	67.0		
	5%	WB	56.4	57.1	60.3	63.2	68.4	74.4	78.0	78.9	75.8	71.5	63.8	58.0		
		MCDB	63.0	63.6	66.6	68.9	74.0	80.4	83.9	85.2	82.0	78.0	70.3	64.8		
	10%	WB	54.7	55.5	59.0	62.0	67.1	72.8	76.8	77.8	74.5	69.7	62.0	56.4		
		MCDB	60.8	61.4	64.9	67.4	72.5	78.7	83.0	84.2	80.8	76.4	68.7	62.8		
(35) Mean Daily Temperature Range	MDBR		15.1	15.7	16.4	16.0	15.0	14.8	14.7	15.1	15.3	16.1	15.8	15.3		
	5% DB	MCDBR	18.9	20.1	20.5	19.6	17.9	18.5	16.9	17.2	16.5	19.3	19.2	18.5		
		MCWBR	11.3	11.8	11.4	11.2	10.4	11.4	11.6	12.2	10.0	11.4	10.9	10.8		
	5% WB	MCDBR	17.9	18.3	18.1	16.8	15.7	16.0	15.2	15.8	15.0	17.1	17.5	17.2		
		MCWBR	11.5	11.7	11.2	10.9	10.3	11.3	11.5	12.1	10.1	11.2	11.1	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.344	0.364	0.422	0.458	0.490	0.491	0.539	0.511	0.483	0.433	0.385	0.350		
	taud		2.467	2.402	2.213	2.132	2.075	2.098	1.973	2.051	2.137	2.257	2.356	2.458		
	Ebn at Noon		267	275	266	263	255	254	241	245	245	246	247	256		
	Edh at Noon		27	32	42	48	52	51	57	52	45	37	30	26		
(44) All-Sky Solar Radiation	RadAvg		746	984	1387	1726	2020	2351	2371	2093	1597	1202	812	673		
	RadStd		82	111	129	142	155	83	62	68	86	80	62	69		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+1.22	N/A	+0.77	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page